Grade: Check Plus

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Assignment #2

Price (x)	f(x)	x f(x)	$(x-25.41)^2 f(x)$
20	.05	1	1.46
21	.06	1.26	1.17
22	.07	1.54	0.81
23	.08	1.84	0.46
24	.09	2.16	0.18
25	.12	3	0.02
26	.15	3.9	0.05
27	.13	3.51	0.33
28	.10	2.8	0.67
29	.10	2.9	1.29
30	.05	1.5	1.05
		25.41	$7.50 = \sigma^2$

$\sigma = 2.74$

1.) mean = \$25.41 ✓ Q1

Standard dev. = 2.74

2.a.) $P(25.41 - 2.74 \le X \le 25.41 + 2.74)$

$P(22.67 \le X \le 28.15)$

$P(23 \le X \le 28) = f(23) + f(24) + f(25) + f(26) + f(27) + f(28)$ ✓ Q2a

$= 0.67$ or 67%

b.) $P(25.41 - 2(2.74) \le X \le 25.41 + 2(2.74))$

$P(19.93 \le X \le 30.89)$

$P(20 \le X \le 30) = 1$ or 100%

✓ Q2b

3.)

Price (x)	f(x)	Option Value	Probability of Option Value
20	.05	0	0.47
21	.06	0	
22	.07	0	
23	.08	0	
24	.09	0	
25	.12	0	
26	.15	1	0.15
27	.13	2	0.13
28	.10	3	0.10
29	.10	4	0.10
30	.05	5	0.05

✓ Q3

4.) $P(X > 0) = 1 - P(0) = 0.53$ or 53%

✓ Q4

5.)

Price (x)	f(x)	Option Value	G(x)	VG(x)	
20	.05	0	0.47	0	0.87
21	.06	0			
22	.07	0			
23	.08	0			
24	.09	0			
25	.12	0			
26	.15	1	0.15	0.15	0.02
27	.13	2	0.13	0.26	0.05
28	.10	3	0.10	0.30	0.27
29	.10	4	0.10	0.40	0.70
30	.05	5	0.05	0.25	0.66
				1.36	$2.57 = \sigma^2$

mean = \$1.36 ✓ Q5

Variance = 2.57

Standard dev. = $\sqrt{2.57}$ = \$1.60